

BINARY ARITHMETIC PRACTICE SHEET

NIMCET TARGET PAPER • EASY LEVEL (SET 2)

ResourceOS

1. What is the binary sum of 1001_2 and 0110_2 ?

- (A) 1111
(C) 1101

- (B) 1110
(D) 1011

2. Subtract 0011_2 from 1010_2 in binary arithmetic.

- (A) 0111
(C) 1001

- (B) 0110
(D) 0101

3. What is the product of binary numbers 110_2 and 10_2 ?

- (A) 1100
(C) 1110

- (B) 1010
(D) 1000

4. Divide 10100_2 by 0101_2 .

- (A) 100
(C) 101

- (B) 011
(D) 010

5. What is the 1's complement of the binary string 11001010?

- (A) 00110101
(C) 11001011

- (B) 00110110
(D) 01001010

6. What is the 2's complement of 01010101?

- (A) 10101011
(C) 10101100

- (B) 10101010
(D) 01010110

7. Convert decimal -12 into 8-bit 2's complement format.

- (A) 11110100
(C) 10001100

- (B) 11110011
(D) 11111100

8. In an 8-bit 2's complement representation, what signed decimal value is 11111010?

- (A) -6
(C) -250

- (B) -10
(D) +250

9. In basic binary addition, what is the result of $1 + 1$?

- (A) Sum = 0, Carry = 1
(C) Sum = 1, Carry = 1

- (B) Sum = 1, Carry = 0
(D) Sum = 2, Carry = 0

10. Perform binary subtraction: $11011_2 - 10100_2$.

- (A) 00111
- (C) 01001

- (B) 00110
- (D) 00101

11. What is the binary product of 1011_2 and 101_2 ?

- (A) 110111
- (C) 111001

- (B) 101111
- (D) 100111

12. What is the quotient when 11110_2 is divided by 110_2 ?

- (A) 101
- (C) 011

- (B) 110
- (D) 100

13. What is the 2's complement of the bit sequence 10000000 in an 8-bit system?

- (A) 10000000
- (C) 00000000

- (B) 01111111
- (D) 11111111

14. To negate a signed number in 2's complement system, one must:

- (A) Invert all bits and add 1
- (C) Add 1 to the MSB only

- (B) Invert all bits and subtract 1
- (D) Shift left by 1 bit

15. Compute the sum of $0101_2 + 0011_2 + 0100_2$.

- (A) 1100
- (C) 1110

- (B) 1010
- (D) 1011

16. Find the 8-bit 1's complement representation of decimal 20.

- (A) 11101011
- (C) 11101100

- (B) 00010100
- (D) 11110100

17. What is the 8-bit 2's complement representation of decimal -2?

- (A) 11111110
- (C) 10000010

- (B) 11111101
- (D) 11111111

18. Divide 100000_2 by 1000_2 .

- (A) 100
- (C) 1000

- (B) 010
- (D) 011

19. Multiply 1001_2 by 11_2 .

- (A) 11011 (B) 10101
(C) 11101 (D) 10011

20. If you add 1 to the 1's complement of any binary number N , the resulting value is:

- (A) 2's complement of N (B) Original number N
(C) 0 (D) $-N$ in sign-magnitude

21. What is the largest positive integer that can be stored in an 8-bit signed 2's complement format?

- (A) +127 (B) +128
(C) +255 (D) +63

22. Perform 8-bit 2's complement addition: $00001010 (+10) + 11110110 (-10)$.

- (A) 00000000 (carry discarded) (B) 11111111
(C) 00010100 (D) 11101100

23. Performing a logical right shift by 2 bit positions ($1100_2 \gg 2$) yields:

- (A) 0011 (B) 0110
(C) 1001 (D) 1111

24. What is the value of $11110_2 - 01010_2$ expressed in decimal?

- (A) 20 (B) 15
(C) 18 (D) 22

25. In binary logic circuits, the sum bit of a Half Adder performing addition of two single bits is generated using which logic gate?

- (A) XOR gate (B) AND gate
(C) OR gate (D) NAND gate

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)